Use Case Diagram Analysis

for

MomentsHub: A Collaborative Daily Photo Capsule

Submitted by:

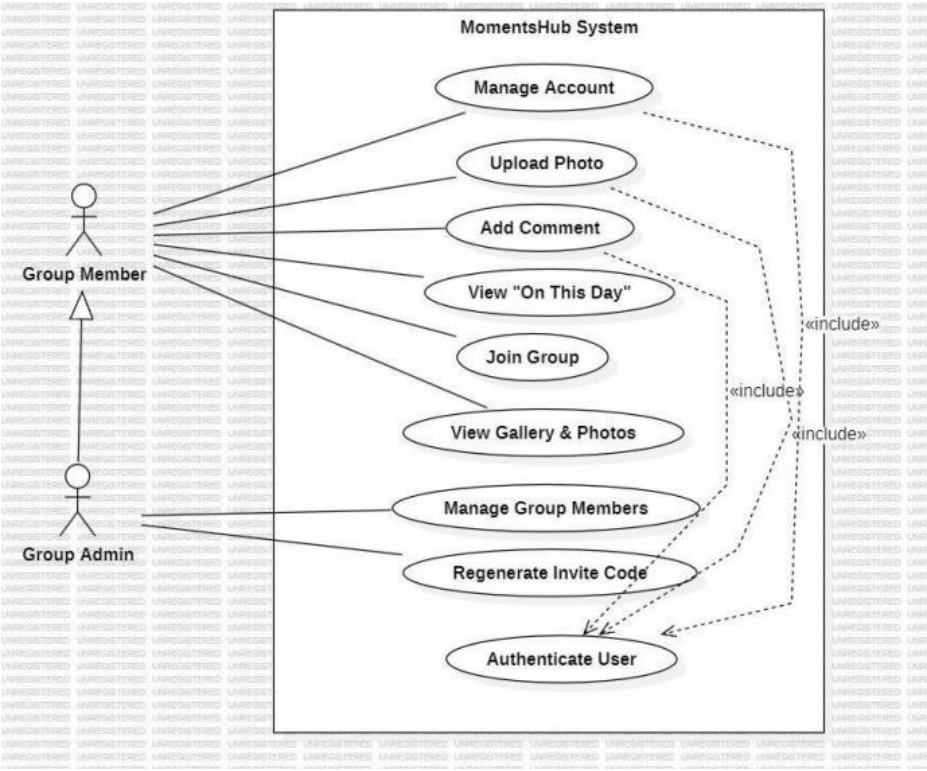
Name: Sasmith.G **Reg. No:** 23BPS1074

1. Introduction

A Use Case Diagram is a behavioral diagram in the Unified Modeling Language (UML) that provides a high-level, graphical overview of a system's functionality. It captures the dynamic behavior of a system by showing the interactions between external entities, called "actors," and the functions they can perform, called "use cases."

The purpose of this document is to present and describe the Use Case Diagram for the **MomentsHub** project. This diagram defines the scope of the system, identifies the key user roles, and outlines the primary functionalities that will be available in the application. It serves as a crucial tool for understanding the system's requirements from a user's perspective.

2. MomentsHub Use Case Diagram



3. Diagram Components

The diagram is composed of three main types of components: Actors, Use Cases, and the relationships that connect them.

3.1. Actors

Actors are external entities that interact with the system. For MomentsHub, we have identified two primary actors:

- **Group Member:** This actor represents a standard user of the application. They have access to all the core social and content-sharing functionalities. Their primary goal is to share and interact with photos within their private groups.
- **Group Admin:** This actor is a specialized type of `Group Member`. A Group Admin inherits all the capabilities of a standard member but possesses additional privileges for managing a group they have created.

3.2. Use Cases

Use cases represent the specific functionalities or services that the system provides to the actors.

Core Use Cases (Available to Group Member)

- **Manage Account:** Allows a user to create their profile, update their display name and avatar, and manage their settings.
- **Upload Photo:** The core function of adding a daily photo to a group.
- **Add Comment:** Allows a user to post text-based comments on a photo.
- **View "On This Day":** Enables a user to see photos shared on the same date in previous years.
- **Join Group:** Allows a user to become a member of a group using an invite code.
- **View Gallery & Photos:** The fundamental ability to browse all photos within a group.

Admin-Specific Use Cases (Available to Group Admin)

- **Manage Group Members:** Allows the admin to remove members from their group.
- **Regenerate Invite Code:** Provides the admin with the ability to change the group's invite code for security or management purposes.

System-Level Use Cases

- **Authenticate User:** A crucial internal use case that handles the login and verification of a user's identity before granting access to protected functionalities.

3.3. Relationships

Relationships define how the actors and use cases interact with each other.

- **Association:** A solid line connecting an actor to a use case. It indicates that the actor can initiate and participate in that use case. For example, the `Group Member` is associated with the `Upload Photo` use case.
- **Generalization (Inheritance):** A solid line with a hollow, triangular arrowhead. In this diagram, it points from the `Group Admin` to the `Group Member`. This signifies that the `Group Admin` is a specialized type of `Group Member` and inherits all of its associated use cases.
- **Include (`<<include>>`):** A dashed arrow that shows a dependency between two use cases. The arrow points from the base use case to the included use case. In this diagram, it signifies that use cases like `Manage Account`, `Upload Photo`, and `Add Comment` require the `Authenticate User` use case to be completed first. This ensures that only logged-in users can perform these actions.